

# Jesús Díaz Pereira

Biomedical Engineer | MSc Candidate in Human-Centered AI

*Specializing in Deep Learning for Neuroimaging*

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## Profile

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Biomedical Engineer and MSc candidate in Human-Centered Artificial Intelligence at DTU, specializing in deep learning for MRI reconstruction and artifact correction. My work focuses on artifact correction to allow reliable quantitative microstructural mapping at 7T. I bridge complex MR physics with AI solutions and have experience in large-scale neuroimaging pipelines and multi-center harmonization for clinical research.

## Professional Experience

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**Student Engineer – MR Science Team** | *TracInnovations, Denmark*      Sept. 2024 – Present

- Led the design and execution of fMRI studies evaluating prospective motion correction performance, generating evidence that supported external dissemination at various conferences.
- Contributed to the development and technical validation of a k-space-domain retrospective motion correction pipeline using markerless optical tracking-derived motion parameters, enabling downstream translation into commercial software.
- Partnered with MRI vendors to evaluate integration pathways for motion correction within scanner reconstruction workflows, resulting in a co-authored ISMRM abstract with Siemens Healthineers.

**Student Assistant – Neuroimaging in MS Group** | *DRCMR, Denmark*      Sept. 2023 – Aug. 2024

- Developed and maintained preprocessing pipelines for ultra-high-field dMRI and qMRI, producing analysis-ready datasets for Multiple Sclerosis research studies.
- Harmonized neuroimaging datasets to be shared with the MAGNIMS network, supporting multi-center analysis of 7T imaging data.

**Research Assistant – Medical Image Analysis Lab** | *Spain*      Nov. 2022 – Jun. 2023

- Led diffusion MRI analysis and statistical evaluation for an industry-partnered study on post-COVID-19 anosmia to investigate neuroplasticity following olfactory rehabilitation.
- Produced imaging-derived metrics and quantitative analyses that supported dissemination as a digital poster at ESMRMB 2023; manuscript in preparation.
- Work supported by a National Research Collaboration Scholarship from the Spanish Ministry of Education.

## Education

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**Technical University of Denmark (DTU)**      2023 – 2026 (Expected)  
MSc in Human-Centered Artificial Intelligence

**Universidad Rey Juan Carlos, Spain**      2019 – 2023  
BSc in Biomedical Engineering (6 courses passed with honors).

## Technical Skills

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|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Computational Methods:</b>            | Deep learning for inverse problems, 3D deformable image registration, domain adaptation, harmonization, statistical modeling, longitudinal analysis |
| <b>Programming &amp; Infrastructure:</b> | Python, Torch, MATLAB, Git, Linux, HPC clusters, pipeline automation, scientific programming                                                        |
| <b>Neuroimaging &amp; MRI:</b>           | Diffusion MRI, functional MRI, 7T UHF imaging, FreeSurfer, FSL, SPM, MRtrix3                                                                        |

## Research Projects

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### Deep Learning for Susceptibility and Eddy Current Distortion Correction in UHF Diffusion MRI

*MSc Thesis – DTU and Danish Research Centre for Magnetic Resonance (2025–2026)*

- Developing a physics-informed deep learning model for 3D deformable registration for joint susceptibility and eddy current distortion correction in UHF diffusion MRI data.

### Analysis of Olfactory Rehabilitation in Post-COVID-19 Anosmia Patients using Diffusion MRI

*BSc Thesis – URJC and GE Healthcare (2022–2023)*

- Conducted diffusion MRI longitudinal study to assess neuroplasticity following olfactory rehabilitation in patients with olfactory impairment due to long COVID.

## Conference Abstracts & Posters

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**ISMRM 2026** | *Harnessing Open Innovation for Integrated Retrospective Motion Correction* Cape Town, South Africa

*Coauthor.* Integrating retrospective motion correction from external tracker into Siemens OpenRecon. To be presented as Digital Poster.

**Evaluation of Prospective Motion Correction in Functional MRI Using Markerless Optical Tracking.** *ISMRM Annual Meeting & Exhibition*, Honolulu, HI, USA, 2025. First Author. Digital Poster

**Diffusion MRI Tractography Reveals Neuroplasticity Following Olfactory Rehabilitation in Post-COVID-19 Anosmia.** *ESMRMB Annual Scientific Meeting*, Basel, Switzerland, 2023. Coauthor. Digital Poster

## Honors & Scholarships

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**Research Collaboration Scholarship** | *Spanish Ministry of Education* 2022 – 2023

- National scholarship awarded to 2,000 students annually to initiate research careers. Integrated into the Department of Applied Mathematics and Materials Science (URJC) to develop the project: *Analysis of olfactory rehabilitation in COVID-19 using MR techniques.*

## Languages

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**Spanish:** Native

**English:** Full Professional Proficiency (CAE – C1)